

Homework Submission

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Problem Statement

Evaluate the indefinite double integral

$$\iint (x^2 + 1) \, dx \, dy.$$

Key Principles and Formulas

- **Iterated integration:** An indefinite double integral can be computed by integrating successively with respect to each variable.
- **Antiderivative of a polynomial:**

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1).$$

- **Constant of integration:** After each integration a "constant" may depend on the remaining variable(s).

Detailed Step-by-Step Solution

First integrate with respect to x (treating y as a constant).

$$\begin{aligned} \int (x^2 + 1) \, dx &= \int x^2 \, dx + \int 1 \, dx \\ &= \frac{x^3}{3} + x + C_1(y), \end{aligned}$$

where $C_1(y)$ is an arbitrary function of y .

Now integrate the result with respect to y .

$$\begin{aligned} \int \left(\frac{x^3}{3} + x + C_1(y) \right) dy &= \left(\frac{x^3}{3} + x \right) \int dy + \int C_1(y) \, dy \\ &= \left(\frac{x^3}{3} + x \right) y + C_2(x), \end{aligned}$$

where $C_2(x) = \int C_1(y) dy$ is an arbitrary function of x . Since the sum of an arbitrary function of x and an arbitrary constant can be absorbed into a single overall constant, we write the most general antiderivative as

$$F(x, y) = \left(\frac{x^3}{3} + x \right) y + C,$$

with C a constant (independent of both x and y).

Final Answer

$$\boxed{\iint (x^2 + 1) dx dy = \left(\frac{x^3}{3} + x \right) y + C}$$

Study Notes and Key Takeaways

- When performing an indefinite double integral, treat the outer variable as a constant during the inner integration; the "constant of integration" may be a function of the outer variable.
- After the second integration, any remaining function of the first variable can be merged into a single overall constant, because the derivative of a constant (with respect to either variable) is zero.
- A common pitfall is to forget the functional "constant" after the first integration; always include $C_1(y)$ (or the appropriate variable) before proceeding to the next step.